

Assignment: Probability and Statistical Modeling of Web Traffic

Using the WorldCup98 Web Server Logs Dataset

Course Topics	Chapters 1-4	Recommended Tool	Python / Jupyter Notebook
Dataset	WorldCup98 Web Server Logs	Deliverables	Notebook + short report

Scenario. You are a performance engineer analyzing real web traffic from the 1998 FIFA World Cup website. Your goal is to use probability and statistical models to understand request arrivals, request behavior, and the variability of web traffic.

Dataset

WorldCup98 Web Server Logs. This public dataset contains real HTTP request logs from the 1998 FIFA World Cup website. It is especially useful for studying request arrivals, interarrival times, object sizes, request categories, and event probabilities. Because it is a workload/access-log dataset rather than a latency benchmark, this assignment focuses on arrival behavior and request characteristics rather than true server response time.

Recommended Dataset Access (Zenodo)

Use the three-day WorldCup98 subset hosted on Zenodo for this assignment. It contains three compressed access-log files (38.9 MB total) and is much more manageable than the complete archive. Dataset page: [1998 World Cup Website Access Logs - Zenodo](#)

- **TR1.gz** - access traffic from May 20, 1998 (17.7 MB)
- **TR2.gz** - access traffic from May 4, 1998 (7.2 MB)
- **TR3.gz** - access traffic from May 23, 1998 (14.0 MB)

DOI: [10.5281/zenodo.5145855](#). The files are gzip-compressed binary logs. You should document any preprocessing or conversion used before analysis.

Data Preprocessing and Starter Code

The Zenodo TR1.gz, TR2.gz, and TR3.gz files are gzip-compressed binary logs, not CSV or plain-text web logs. After decompression, each request is stored as a fixed 20-byte binary record. Use the starter code below to convert the records into a Pandas DataFrame before beginning the statistical analysis.

Binary Record Structure

Field	Data type	Size
timestamp	uint32	4 bytes
client_id	uint32	4 bytes
object_id	uint32	4 bytes
size	uint32	4 bytes
method	uint8	1 byte
status	uint8	1 byte
type	uint8	1 byte
server	uint8	1 byte

The four 32-bit integers are stored in big-endian (network) byte order, so the Python struct format used below begins with >.

Required Preprocessing

1. Read and decompress one of the .gz files directly in Python.
2. Unpack each 20-byte request record.
3. Create a Pandas DataFrame containing timestamp, client_id, object_id, size, method, status, type, and server.
4. Convert timestamp to a readable datetime.
5. Sort requests by timestamp.
6. Create derived variables needed for the statistical analysis, especially interarrival_time and request counts per fixed time interval.

Starter Python Code

```
import gzip
import struct
import pandas as pd

records = []

with gzip.open("TR1.gz", "rb") as f:
    #with open(file_path, "rb") as f: #if it is autoDecompressed in your MACOS
        while True:
            data = f.read(20)
            if len(data) < 20:
                break

            timestamp, client, obj, size, method, status, file_type, server = \
                struct.unpack(">IIIBBBB", data)

            records.append([
                timestamp, client, obj, size,
                method, status, file_type, server
            ])
    ])
```

```
df = pd.DataFrame(records, columns=[
    "timestamp", "client_id", "object_id", "size",
    "method", "status", "type", "server"
])

df["datetime"] = pd.to_datetime(df["timestamp"], unit="s", utc=True)
df = df.sort_values("timestamp").reset_index(drop=True)

print(df.head())
print(df.shape)
```

Create Variables for the Assignment

Interarrival time between consecutive requests:

```
df["interarrival_time"] = df["timestamp"].diff()
```

Number of requests per minute:

```
df["minute"] = df["datetime"].dt.floor("min")
requests_per_minute = df.groupby("minute").size().reset_index(name="request_count")
```

These derived variables connect directly to the probability models in the assignment: request_count can be studied with a Poisson model, interarrival_time with Exponential models, and size with distributions such as Lognormal.

Scope of Preprocessing

For this statistics assignment, you are not required to reverse-engineer every packed field. The timestamp, client_id, object_id, size, and type fields are sufficient for most tasks. Decoding the detailed status and server bit fields is optional unless you choose to use them in your analysis.

You may use TR1.gz alone for a smaller analysis, or combine TR1.gz, TR2.gz, and TR3.gz after preprocessing. Clearly state which file(s) you used in your notebook and report.

Learning Objectives

- Define sample spaces, events, random variables, and probability models from observed web traffic.
- Use conditional probability, independence, total probability, and Bayes' theorem.
- Build and interpret PMFs, CDFs, means, and variances for discrete random variables.
- Evaluate Binomial, Geometric, Negative Binomial, Hypergeometric, and Poisson models.
- Evaluate continuous probability models for interarrival time and object size, including Exponential, Normal, and Uniform distributions.
- Compare empirical behavior with theoretical probability models and explain model limitations.

Part 1 - Probability from Observed Events

Define two events from the data. For example:

- $A = \{\text{request belongs to a selected request category}\}$
 - $B = \{\text{object size exceeds a chosen threshold}\}$
1. Estimate $P(A)$, $P(B)$, $P(A \cap B)$, and $P(A \cup B)$. Verify the addition rule numerically.
 2. Estimate $P(A | B)$ and $P(B | A)$. Interpret both probabilities in the context of web requests.

3. Investigate whether A and B appear independent. Support your conclusion with numerical evidence.
4. Partition requests into several request categories and use the law of total probability to compute the probability of a selected event.
5. Use Bayes' theorem to compute the probability that a request came from a particular category given that another event occurred. Compare the theoretical result with the corresponding proportion measured directly from the dataset.

Part 2 - Discrete Random Variables and Distributions

Create equal time intervals (for example, 1-minute intervals) and define **X = number of requests arriving in one interval**.

6. Construct the empirical probability mass function (PMF) of X.
7. Construct the empirical CDF of X and use it to estimate probabilities such as $P(X \leq x)$ and $P(a < X \leq b)$.
8. Calculate the sample mean and variance of X.
9. Fit a Poisson model to X by estimating λ from the data. Compare the observed mean, observed variance, and empirical frequencies with the theoretical Poisson model.
10. Define a binary request event and estimate its probability p. For groups of n requests, use a Binomial model to calculate the probability of exactly k events, no events, and at least one event.
11. Define Y as the number of requests until the first occurrence of your selected event. Use a Geometric distribution to model Y and compare the theoretical and empirical means.
12. Extend the previous question to the number of requests required to observe the third occurrence of the event. Identify the appropriate distribution and calculate its expected value.

Part 3 - Continuous Random Variables and Distributions

Define **T = interarrival time between consecutive requests**.

13. Calculate the mean, variance, standard deviation, median, and selected percentiles of T. Plot a histogram and empirical CDF.
14. Compare the interarrival-time data with at least three continuous distributions, including Exponential and Normal.
15. Estimate the required parameters for each model and overlay the theoretical density on the observed histogram.
16. Discuss which distribution appears to describe interarrival time most reasonably. Explain why real web traffic may differ from idealized textbook models. (Bursty traffic, Automated Bots, Attacks, Day vs Night variations)
17. Using your preferred model, estimate probabilities such as $P(T > t_0)$ and $P(t_1 < T < t_2)$. Compare these theoretical probabilities with empirical proportions.
18. Repeat part of the continuous-distribution analysis using object size as the random variable. In particular, investigate whether a Lognormal model is reasonable.
19. Investigate whether a Uniform distribution is reasonable for any continuous variable or transformed subset of the data. Explain your conclusion.

Part 4 - Normal Approximations

20. For a sufficiently large number n of requests, let Z be the number satisfying your selected binary event (success/failure). Calculate $P(Z \geq k)$ using both the exact Binomial distribution and the Normal approximation with continuity correction. Compare the results.

21. For a time interval with sufficiently large expected request count λ , calculate a request-count probability using both the exact Poisson distribution and the Normal approximation. Report the difference.

Final Engineering Analysis

Write a 1-2 page interpretation answering the following question:

What have you learned about the behavior and variability of this web system from the data?

- Identify which probability models fit reasonably well and which do not.
- Explain possible reasons for deviations from theoretical models.
- Distinguish conclusions supported directly by the data from conclusions that depend on modeling assumptions.
- Include at least two meaningful figures and one summary table in your final interpretation.

Deliverables

- One Jupyter Notebook (.ipynb) containing data preparation, calculations, plots, model comparisons, and short interpretations throughout.
- One short report (PDF or DOCX) containing the 1-2 page final engineering analysis and selected figures/tables.
- Code must be reproducible from the given dataset and from clearly documented preprocessing steps.

Suggested Grading

Component	Weight
Probability and conditional probability	30%
Discrete random variables and distributions	30%
Continuous distributions and approximations	30%
presentation	10%

Submission Notes

- Use meaningful variable names and label all axes, tables, and figures.
- Show formulas or identify the probability model used before presenting numerical results.
- Do not report only numerical answers; briefly interpret each result in the context of the web system.
- When a theoretical distribution does not fit the data well, explain why rather than forcing the model.